

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414260
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Lebovitz; Adam Nathan et al.

Controller for unmanned aerial vehicles

Abstract

A controller for an unmanned aerial vehicle may have a top portion configured to interface with a user and a bottom portion on a side opposite the top portion. A device support may be coupled to the controller. The device support may be configured to hold a portable electronic device. The device support may be movable between a closed position in which the device support is received in the bottom portion of the controller and an open position in which the device support extends away from the controller.

Inventors: Lebovitz; Adam Nathan (San Mateo, CA), Thompson; Benjamin Scott (San Carlos, CA), O'Rourke; Kellen James Waterman (Belmont, CA)

Applicant: Skydio, Inc. (Redwood City, CA)

Family ID: 87558055

Assignee: Skydio, Inc. (San Mateo, CA)

Appl. No.: 17/862799

Filed: July 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230261359 A1	Aug. 17, 2023

Related U.S. Application Data

us-provisional-application US 63309848 20220214

Publication Classification

Int. Cl.: H05K7/20 (20060101); G05D1/00 (20240101); G05D1/221 (20240101); H01Q1/12 (20060101); H01Q21/28 (20060101); H04M1/04 (20060101); H04M1/72409 (20210101)

U.S. Cl.:

CPC H05K7/20163 (20130101); G05D1/221 (20240101); H01Q1/1235 (20130101); H04M1/72409 (20210101); H05K7/20136 (20130101); H05K7/2039 (20130101); B64U2201/20 (20230101); G05D1/0011 (20130101); H01Q21/28 (20130101); H04M1/04 (20130101)

Field of Classification Search

CPC: H05K (7/20163); H05K (7/20136); H05K (7/2039); H05K (5/00); H05K (7/00); G05D (1/0011); H04M (1/04); H04M (1/72409); H01Q (1/1235); H01Q (21/28); H01Q (1/22); B64U (2101/30); B64U (2201/20); A63H (30/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6097595	12/1999	Cipolla	248/688	G06F 1/166
11226617	12/2021	Deng	N/A	G05D 1/0016
11420741	12/2021	Deng	N/A	G05D 1/0011
2014/0364232	12/2013	Cramer	463/37	A63F 13/98
2015/0174482	12/2014	Hirshberg	463/37	A63F 13/92
2016/0355263	12/2015	Pozzi	N/A	B64D 11/00152
2017/0265313	12/2016	Deng	N/A	F16C 11/10
2019/0287387	12/2018	Keller	N/A	F16M 11/041
2020/0088344	12/2019	Tsuji	N/A	F16M 11/38
2022/0230936	12/2021	Zhong	N/A	H01L 25/0652
2022/0326705	12/2021	Thompson	N/A	B64C 39/024
2023/0228365	12/2022	Watanabe	248/122.1	F16M 13/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
208300209	12/2017	CN	N/A
112771590	12/2020	CN	G08C 17/02

OTHER PUBLICATIONS

Ricker, Ed, “Experiencing DJI's Smart Controller—What's There to Like?”, Published Feb. 17, 2019, Retrieved Sep. 12, 2024, URL <<https://youtu.be/QkVMHnH8P4Q?t=299>>, Relevant timestamps 4:59-5:00, 7:07-7:09, 8:02-8:25 (Year: 2019). cited by examiner
“CN112771590—DescriptionMachineTranslation.pdf”—Machine Translation of the Description section of CN 112771590 into English, retrieved through Espacenet on Sep. 18, 2024 (Year: 2021). cited by examiner
“CN208300209U—DescriptionMachineTranslation.pdf”—Machine Translation of the Description

Primary Examiner: Robinson; Kito R

Assistant Examiner: Glade; Zachary E. F.

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims priority to and the benefit of U.S. Provisional Patent Application Ser. No. 63/309,848, filed Feb. 14, 2022, the entire disclosure of which is hereby incorporated by reference.

FIELD

(1) This disclosure relates generally to unmanned aerial vehicles (UAVs) and, more specifically, to a controller for a UAV.

BACKGROUND

(2) A UAV is an aircraft that can fly without a human operator onboard. The flight of a UAV may be controlled by a human operator using a wireless controller to communicate with the UAV.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIGS. 1-2 are isometric views of an example of a controller for a UAV in which the controller includes a device support in a closed position.

(3) FIGS. 3-4 are isometric views of an example of a controller for a UAV in which the controller includes a device support, shown holding a portable electronic device, in an open position.

(4) FIG. 5 is an isometric view of an example of a controller having a device support, shown without a portable electronic device, in an open position.

(5) FIG. 6 is a cross section of an isometric view of an example of a controller having a device support, shown without a portable electronic device, in an open position.

(6) FIG. 7 is an isometric view of an example of a controller with antennas for communicating with a UAV.

(7) FIGS. 8-12 are views of an example of a controller with a fan and heatsink portions for cooling circuitry arranged inside.

(8) FIG. 13 is a flowchart of an example of a process for implementing a controller that includes a device support and a support stand.

(9) FIG. 14 is a flowchart of an example of a process for implementing a controller that includes a fan and heatsink portions for cooling circuitry arranged inside.

DETAILED DESCRIPTION OF THE DRAWINGS

(10) To control an unmanned aerial vehicle (UAV), a human operator (also referred to as a “user”) may use a portable electronic device (or simply “device”), such as a smart phone or tablet, in conjunction with a wireless UAV controller (or simply “controller”). The user may provide inputs

to, and may receive outputs from, an application executing on the device. The device may communicate with the controller, and the controller may communicate with a UAV, such as via communications circuitry connected to one or more outwardly extending antennas.

(11) It may be desirable at times to transport and/or store the controller when the controller is not being used. However, the system used to connect the device to the controller, and/or the antennas used by the controller, may make transportation and/or storage difficult. For example, the system used to connect the device to the controller may be bulky or awkward for transportation and/or storage, and the outwardly extending antennas may be susceptible to damage should the controller be dropped. A need therefore exists to improve the system used to connect the device to the controller, and/or to improve the antennas used by the controller, to better facilitate transportation and/or storage.

(12) Additionally, during assembly, it may be desirable to install different communications circuitry in the controller. For example, the communications circuitry used by the controller to communicate with a UAV in a first geographic region (e.g., the United States) may be different than the communications circuitry used by the controller to communicate with a UAV in a second geographic region (e.g., Europe). However, the arrangement of other circuitry in the controller (e.g., power circuitry used to receive and distribute electrical power, and/or processing circuitry used to provide intelligent control of the system), and the arrangement of cooling systems in the controller (e.g., fans and/or heatsinks), may make changing the communications circuitry difficult. Moreover, the limited space available inside the controller, which may be reduced in size to be conveniently held by a user, may make including variations of the communications circuitry impractical. A need therefore exists to improve the assembly of controller to permit the efficient assembly (e.g., the efficient method of assembly) and/or exchange of circuitry arranged inside.

(13) Implementations of this disclosure address problems such as these by coupling a device support (e.g., a rotatable, telescoping arm) to a controller. The device support may be configured to hold a portable electronic device of various sizes, such as a smart phone, mini tablet, or full size tablet. The device support may be movable between a “closed” position in which the device support is received in a bottom portion of the controller (e.g., in a cavity in the bottom portion), such as when the controller is not in use, and an “open” position in which the device support extends away from the controller, such as when the controller is in use. The bottom portion of the controller may be on a side opposite the top portion of the controller. The top portion may be configured to interface with the user for controlling the UAV. For example, the top portion may include control elements accessible to the user, such as buttons, directional pads, light emitting diodes (LEDs), and joysticks (e.g., accessible by a user's thumbs), while the bottom portion may include an ergonomic grip surface for holding the controller (e.g., accommodating a user's hands on opposing sides of the controller **100**). Moving the device support to under the controller when not in use (e.g., folding, swinging, or collapsing underneath) may permit greater ease of transportation and/or storage. Moreover, moving the device support to under the controller may permit the top portion of the controller to be configured for interfacing with a user while reducing undesirable interference with the control elements included in the top portion (e.g., such as an undesirable cavity on the top portion for receiving the device support). In some implementations, a device support hinge may be arranged between the controller and the device support for rotating the device support under the controller (e.g., from the open position to the closed position). In some implementations, one or more antennas for communicating with the UAV may be integrated in the device support. Moving the device support from the closed position to the open position may permit angling the antennas upward and/or outward in a direction overhead toward a UAV.

(14) In some implementations, a support stand may be coupled to the controller. The support stand may be configured to support the controller at an angle relative to a surface, such as when the controller is resting on table, desk, or another platform. The device support may be configured to cause the support stand to be received in the bottom portion of the controller when in the closed

position (e.g., the device support may pull the support stand into a cavity defined by bottom portion). The device support may also be configured to permit the support stand to automatically extend away from the controller (e.g., to support the controller on a platform) when in the open position. In some implementations, one or more antennas for communicating with the UAV may be integrated in the support stand. Moving the device support from the closed position to the open position may permit angling the antennas, integrated in the support stand, outward in a direction overhead toward a UAV when the controller is held by a user and in use. Moving the device support from the open position to the closed position may cause a retraction of the antennas inward when the controller is not in use. Thus, moving the device support from the open position to the closed position may improve protection of the antennas from environmental conditions.

(15) In some implementations, a fan and first and second heatsink portions may be arranged in the controller. The first and second heatsink portions may be configured to cool first and second components arranged in the controller, respectively, such as the first heatsink portion cooling a first component comprising communications circuitry implemented on a communications board (e.g., a first printed circuit board (PCB)) and the second heatsink portion cooling a second component comprising power and/or processing circuitry implemented on a second circuit board (e.g., a second PCB). The first and second heatsink portions may be adjacent to one another, and the fan may generate an airflow between the first and second heatsink portions. In some implementations, fins of the first heatsink portion may be interleaved with fins of the second heatsink portion with air gaps in between. This may permit ease of access to the components, such as the component comprising communications circuitry on the communications board, for changing the components during assembly, without moving other components (e.g., the component comprising power and/or processing circuitry) and/or the cooling system (e.g., the fan or the heatsink portions).

(16) FIGS. 1-2 are isometric views of an example of a controller **100** for a UAV in which the controller **100** includes a device support **110** in a closed position. The controller **100** may be a handheld wireless UAV controller that is configured to be operated by a user to control a UAV. The device support **110**, which may be swingable, foldable, rotatable, and/or telescoping, may receive an attachment or adapter. The device support **110** may be configured to hold a portable electronic device of various sizes, such as different sizes of a smart phone. The attachment or adapter may permit the device support **110** to hold a greater number of sizes for the portable electronic device, such as a mini tablet or full size tablet. The device support **110** may be movable between a “closed” position in which the device support **110** is received in a bottom portion **114** of the controller **100** (e.g., shown in FIG. 2, received in a cavity in the bottom portion **114** to at least partially form an exterior surface of the bottom portion **114**), and an “open” position in which the device support **110** extends away from the controller **100** (e.g., shown in FIGS. 3 and 4). The closed position may be used when the controller **100** is not in use, and the open position may be used when the controller **100** is in use.

(17) The bottom portion **114** of the controller **100** (e.g., shown in FIG. 2) may be on a side opposite a top portion **112** of the controller **100** (e.g., shown in FIG. 1). The top portion **112** may be configured to interface with a user for controlling the UAV. For example, top portion **112** may include control elements **116A-116H** accessible to the user, such as buttons, directional pads, LEDs, and joysticks (e.g., accessible by a user's thumbs), while the bottom portion **114** may include an ergonomic grip surface for holding the controller **100** (e.g., accommodating a user's hands on opposing sides of the controller **100**). In one example, the control elements **116A** and **116B** may comprise joysticks; the control elements **116C**, **116D**, and **116E** may comprise buttons (e.g., for interfacing with an application executing on the device when attached by the device support **110**, and/or a return to home button for the UAV); the control element **116F** may comprise a multi-directional pad; the control element **116G** may comprise LEDs (e.g., status indicators for power); and the control element **116H** may comprise a power button for the controller **100**.

(18) It may be desirable at times to transport and/or store the controller **100** when the controller **100**

is not being used. Moving the device support **110** to under the controller **100** when the controller **100** is not in use (e.g., folding, swinging, or collapsing underneath) may permit greater ease of transportation and/or storage of the controller **100**. Moreover, moving the device support **110** to under the controller **100** may permit the top portion **112** of the controller **100** to be configured for interfacing with a user while reducing undesirable interference with the control elements **116A-116H** included in the top portion **112** (e.g., such as an undesirable cavity on the top portion **112** for receiving the device support **110**).

(19) In some implementations, a device support hinge (e.g., the device support hinge **118**, shown in FIGS. 4-6) may be arranged between the controller **100** and the device support **110** for rotating the device support **110** relative to the controller **100** (e.g., from the open position to the closed position, and from the closed position to the open position). Thus, the closed position may be associated with transportation and/or storage of the controller **100**, and the open position may be associated with use of the controller **100**. In the closed position, the device support **110** may be received in the bottom portion **114** of the controller **100**. For example, the device support **110** may fit in a central cavity of the bottom portion **114** when in the closed position. In the open position, the device support **110** may extend away from the bottom portion **114** of the controller **100**. Moving the device support **110** to under the controller **100** when the controller **100** is not in use may permit greater ease of transportation and/or storage of the controller **100**. Moreover, the top portion **112** of the controller **100** may be configured for interfacing with a user without involving modifications for accommodating the device support **110**.

(20) In some implementations, as may be best seen in FIGS. 2 and 4, a support stand **120** may be coupled to the controller **100**. The support stand **120** may be configured to support the controller **100** at an angle relative to a surface, such as when the controller **100** is resting on table, desk, or another platform. This may permit a user to use the controller **100**, and use a device held by the controller **100**, without the user physically holding the controller **100**. The support stand **120** may comprise individual support legs, such as support legs **122A** and **122B**. The device support **110** may be configured to cause the support stand **120** to be received in the bottom portion **114** of the controller **100** when in the closed position. For example, the device support **110** may pull the support legs **122A** and **122B** of the support stand **120** into side-cavities **154** in the bottom portion **114**. The device support **110** may also be configured to permit the support stand **120** to automatically extend away from the controller **100**, such as to support the controller **100** on a platform when in the open position. For example, the support legs **122A** and **122B** of the support stand **120** may be spring loaded, and the device support **110** may release the support legs **122A** and **122B** from the side-cavities **154** in the bottom portion **114** when the device support **110** is moved to the open position.

(21) In some implementations, as may be best seen in FIG. 1, the device support **110** may be configured to lock into a front portion **124** of the controller **100** when in the closed position. Moving the device support **110** from the closed position to the open position may involve pulling the device support **110** away from the front portion **124** of the controller **100** (e.g., unlocking the device support **110** from the front portion **124**) and rotating the device support **110** about the device support hinge **118** (e.g., extending the device support **110** away from the controller **100** by rotating the device support **110** downward from the controller **100** and then back upward over the controller **100** for using a device attached thereto). In some implementations, the front portion **124** may include a port **126** (e.g., a charging port, such as a universal serial bus (USB) port). The device support **110** may include a channel **128** to permit access to the port **126** when the device support **110** is in the closed position (e.g., locked into the front portion **124**). This may permit charging the controller **100**, and a device attached thereto via the device support **110**, including when the controller **100** is in the closed position and not in use.

(22) In some implementations, the controller **100** may include a global positioning system (GPS) device arranged inside the controller **100**. For example, the GPS device may be in addition to any

GPS device provided by the device **115**. The GPS device may be used to recall a UAV to a precise location corresponding to the controller **100**, such as by a single press of one of the control elements **116C**, **116D**, and **116E** (e.g., a single press of the control element **116D**, which could be a return to home button for the UAV).

(23) FIGS. 3-4 are isometric views of an example of the controller **100** in which the controller **100** includes the device support **110**, shown holding a device **115**. The device **115** may, for example, be a portable electronic device, such as a smart phone. To control a UAV, a user may use the device **115** in conjunction with the controller **100**. In some implementations, the device **115** may have a cable **130** connecting the device **115** to a port **132** on the controller **100** (e.g., a USB port on the bottom portion **114** of the controller **100**). The cable **130**, and thus the port **132**, may be used to provide power to the device **115**, and/or to control an application executing on the device **115** (e.g., such as via the one or more of the control elements **116A-116H**). For example, when in use, the user may provide inputs to, and may be exposed to outputs from, an application executing on the device **115**. The device **115** may communicate with the controller **100**, via the cable **130**, and the controller may communicate with a UAV, such as via communications circuitry connected to one or more outwardly extending antennas.

(24) In some implementations, the device **115** may communicate wirelessly with the controller **100** (e.g., without the cable **130**). In some implementations, the controller **100** may also include a port **134** for providing a video output (e.g., a high-definition multimedia interface (HDMI) port on the bottom portion **114** of the controller **100**, for displaying video captured by the UAV). In some implementations, the device **115** may remain attached to the device support **110**, and the cable **130** may remain connected between the device **115** and the controller **100**, in both the open position and the closed position. For example, the bottom portion **114** of the controller **100** may include recesses for accommodating the device **115** and/or the cable **130** when in the closed position.

(25) With additional reference to FIGS. 5-6 (e.g., showing views of the controller **100** in the open position and without the device **115**), the controller **100** may include the device support hinge **118** arranged between the controller **100** and the device support **110**. The device support hinge **118** may permit rotating the device support **110** from under the controller **100** (e.g., from the closed position, as shown in FIGS. 1-2) to over the controller **100** (e.g., to the open position, as shown in FIGS. 3-4), such as for when the controller **100** is in use. The device support hinge **118** may also permit rotating the device support **110** back from over the controller **100** (e.g., from the open position, as shown in FIGS. 3-4) to under the controller **100** (e.g., to the closed position, as shown in FIGS. 1-2), such as for when the controller **100** is not in use. The bottom portion **114** of the controller **100** may include a central cavity **140** for receiving the device support **110** when the device support **110** is in the closed position.

(26) In some implementations, as may be best seen in FIG. 6, the device support **110** may include a lower clamp portion **142**, an upper clamp portion **144**, and an arm **146** in between. The lower clamp portion **142** and the upper clamp portion **144** may be used to clamp (e.g., to securely hold or attach, such as with tension) the device **115** to the device support **110**. In some implementations, the lower clamp portion **142** may be stationary on a lower portion of the arm **146** (e.g., proximal to the controller **100**), while the upper clamp portion **144** may be moveable along an upper portion of the arm **146** (e.g., distal to the controller **100**). In some implementations, the arm **146** may be a spring loaded telescoping arm to permit holding different sizes of a portable electronic device (e.g., different sizes of a smart phone) between the lower clamp portion **142** and the upper clamp portion **144**. For example, the upper clamp portion **144** may be pulled away from the lower clamp portion **142** along the arm **146**, working against a spring, so that the device **115** may be installed between the upper clamp portion **144** and the lower clamp portion **142**. A spring may pull the upper clamp portion **144** and the lower clamp portion **142** back together, against the device **115**, to hold the device **115** in position. Thus, the device support **110** may be extendable to multiple positions for holding different sizes of the device **115**. As may be seen in FIG. 2, the lower clamp portion **142**

and the upper clamp portion **144** may be external to the controller when the device support **110** is in the closed position. In some cases, this may permit the device support **110** to continue to hold the device **115** when the device support **110** is in the closed position.

(27) In some implementations, and as may be best seen in FIG. 2, the device support **110** may also include a device mounting system **148** for securely attaching the device **115** when a larger portable electronic device is used. For example, the device mounting system **148** may comprise a recess for receiving an adapter for holding a larger portable electronic device (e.g., larger than a smartphone), such as a mini tablet or full size tablet (e.g., including sizes up to 10 inches in diameter). In some implementations, the device **115** may remain securely attached to the device support **110**, even when the controller **100** is in the closed position (e.g., in addition to being securely attached when the controller **100** is in the open position). In some implementations, an attachment or adapter may be added to the device support **110**, so that the device support **110** may hold a greater number of sizes for the device **115**.

(28) In some implementations, the device support **110** may be rigidly locked into place when fully extended in the open position. For example, and as may be best seen in FIG. 6, showing a cross section A-B from FIG. 5, moving the device support **110** to the open position may actuate a locking system **150** that is configured to lock the device support **110** in the open position when the device support **110** is fully extended. The locking system **150** may engage with a rotatable feature of the device support hinge **118** to permit moving to the open position and locking in the open position. A release **153** (e.g., a mechanical, slidable button), which may be arranged above the locking system **150**, may be configured to release the device support **110** from the open position (e.g., unlock the device support **110** from the open position), so that the device support **110** can freely rotate back to the closed position. In some implementations, the release **153** may actuate the locking system **150** (e.g., pull the locking system **150** rearward), so that the locking system **150** disengages with the device support hinge **118** to permit the device support hinge **118** to freely rotate the device support **110** back to the closed position.

(29) In some implementations, the support stand **120** may be configured to support the controller **100** at an angle relative to a surface, such as when the controller **100** is resting on table, desk, or another platform. This may permit a user to use the controller **100**, and thus the device **115**, without the user physically holding the controller **100**. The device support **110** may be configured to cause the support stand **120** to also be received in the bottom portion **114** of the controller **100** when in the closed position. For example, as may be best seen in FIG. 5, tabs **151** of the device support **110** may engage with grooves **152** (e.g., pockets) of the support legs **122A** and **122B** to pull the support legs **122A** and **122B** into side-cavities **154** in the bottom portion **114**. The support legs **122A** and **122B** may rotate via a support stand hinge **158**, and the device support **110** may be used to control such rotation. The device support **110** may also be configured to permit the support stand **120** to automatically extend away from the controller **100**, such as to support the controller **100** on a platform when in the open position. For example, when moving the device support **110** to the open position, the tabs **151** of the device support **110** may release from the grooves **152** of the support legs **122A** and **122B** to permit the support legs **122A** and **122B** to resiliently extend from the side side-cavities **154** back outward to the open position (e.g., rotating about the support stand hinge **158**). For example, the support legs **122A** and **122B** may be spring loaded, so that the support legs **122A** and **122B** may move by default to the open position, away from the side side-cavities **154**.

(30) FIG. 7 is an isometric view of an example of the controller **100** with antennas for communicating with a UAV. The device support **110** and the support stand **120** are shown in the open position and without a device attached (e.g., without the device **115**). The controller **100** may include one or more antennas for communicating with the UAV, such as antennas **160A-160D**. The antennas **160A-160D** may be connected to communications circuitry in the controller **100** for communicating with a UAV. For example, the antennas **160A-160D** may be connected to a component **176** (e.g., shown in FIGS. 9 and 10) comprising communications circuitry that is

arranged inside the controller **100**.

(31) The antennas may be integrated in the device support **110** and/or the support stand **120**. For example, the antennas **160A** and **160B** may be integrated in the device support **110**, such as in the upper clamp portion **144**. Moving the device support **110** from the closed position to the open position may permit angling the antennas **160A** and **160B** upward and/or outward in a direction overhead toward a UAV. Also, the antennas **160C** and **160D** may be integrated in the support stand **120**, such as in the support legs **122A** and **122B**, respectively. Moving the device support **110** from the closed position to the open position may also permit angling the antennas **160C** and **160D**, integrated in the support stand **120**, outward in a direction overhead toward a UAV when the controller **100** is held by a user. Additionally, moving the device support from the open position to the closed position may improve protection of the antennas **160A-160D** from environmental conditions, such as by pulling and/or locking the antennas **160A** and **160B** (e.g., integrated in the device support **110**) and the antennas **160C** and **160D** (e.g., integrated in the support stand **120**) in the bottom portion **114** of the controller **100**. Further, the antennas **160A-160D** may be extended away from the controller **100** when in the open position, into a fixed, known orientation, so that a user's hands do not contact the antennas **160A-160D** or otherwise interfere with the antennas **160A-160D**. Also, the antennas **160A-160D** may be extended with horizontal separation (e.g., a horizontal separation **161A**, between the antennas **160A** and **160B** in the device support **110**, and a horizontal separation **161B**, between the antennas **160C** and **160D** in the support stand **120**) and with vertical separation (between the antennas **160A** and **160B** in the device support **110**, arranged upward, and the antennas **160C** and **160D** in the support stand **120**, arranged downward) for improved communication with a UAV.

(32) In some implementations, the antennas **160A-160D** may be used to communicate via different frequency bands. For example, the antennas **160A** and **160B** in the device support **110** may be used to communicate via 2.4 GHz Wi-Fi (e.g., based on the IEEE 802.11 family of standards), and the antennas **160C** and **160D** in the support stand **120** may be used to communicate via 5 GHz Wi-Fi.

(33) FIGS. **8-12** are views of an example of the controller **100** with a fan **170** (e.g., a blower fan) and heatsink portions, such as heatsink portions **172** and **174**, arranged inside the controller **100**. During assembly of the controller **100**, it may be desirable to install different communications circuitry in the controller **100**. For example, as may be best seen in FIGS. **9** and **10**, the component **176** may comprise communications circuitry used by the controller **100** to communicate with a UAV in a first geographic region (e.g., the United States). It may be desirable to exchange the component **176** with a component comprising different communications circuitry to communicate with a UAV in a second geographic region (e.g., Europe). However, the arrangement of other circuitry in the controller, such as a component **178** comprising power and/or processing circuitry (e.g., power circuitry used to receive and distribute electrical power in the controller **100**, processing circuitry used to provide intelligent control of the controller **100**, and/or a GPS device used to recall a UAV to a precise location of the controller **100**), may make changing the component **176** difficult. Moreover, the limited space available inside the controller **100**, which may be reduced in size to be conveniently held by a user, may make including variations of the components (e.g., variations of communications circuitry) impractical. Thus, the fan **170** and the heatsink portions **172** and **174** may be configured in the controller **100** to permit ease of access to the component **176**, such as for changing the communications circuitry.

(34) In some implementations, the heatsink portions **172** and **174** may comprise a single heatsink. For example, the single heatsink may have an open channel in the middle to permit an airflow to pass therethrough. In some implementations, the heatsink portions **172** and **174** may comprise multiple heatsinks. For example, the multiple heatsinks may be physically joined on sides with an open channel in the middle to permit an airflow to pass therethrough. In some implementations, the heatsink portions **172** and **174** may include thermal interface material (e.g., between a heatsink portion and a component). In some implementations, the heatsink portions **172** and **174** may

include fins for dissipating heat.

(35) As may be best seen in FIG. 9, the heatsink portions **172** and **174** may be configured to cool circuitry arranged in the controller **100** in different planes, such as the heatsink portion **172** cooling the component **176** (e.g., comprising the communications circuitry) in an upper plane, and the heatsink portion **174** cooling the component **178** (e.g., comprising the power and/or processing circuitry) in a lower plane. The heatsink portion **172** may cool the component **176** implemented on a communications board **180** in the upper plane (e.g., a radio board, which may be secondary PCB and which may be wired to antennas such as the antennas **160A-160D**), and the heatsink portion **174** may cool the component **178** implemented on a power and/or processing board **182** in the lower plane (e.g., a main board, which maybe a primary PCB and which may be wired to ports such as the port **126**, the port **132**, and the port **134**). The heatsink portions **172** and **174** may be mounted to the boards, such as via fasteners (e.g., screws and/or retaining clips). For example, the heatsink portion **172** may be mounted to the communications board **180**, and the heatsink portion **174** may be mounted to the power and/or processing board **182**, via one or more fasteners.

(36) The heatsink portions **172** and **174** may be adjacent to one another, and the fan **170** may generate an airflow between the heatsink portions **172** and **174** adjacent to one another. For example, as may be best seen in FIG. 11, the airflow may ingress from an inlet **184** arranged in the bottom portion **114** of the controller **100**, and the inlet **184** may be exposed to ambient air when in the open position. Also, as may be best seen in FIG. 12, the airflow may egress (e.g., exhaust) at an outlet **186** arranged in the front portion **124** of the controller **100** that remains at least partially unobstructed by the controller **100** or components thereof, which may also be exposed to ambient air when in the open position. The fan **170** may generate the airflow from the inlet **184**, turn the airflow 90 degrees, force the airflow between the heatsink portions **172** and **174**, and exhaust the airflow at the outlet **186**, to move heated air out of the controller **100**. For example, the fan **170** and the heatsink portions **172** and **174** may use forced convection to dissipate heat from the components **176** and **178**. The fan **170** may generate the airflow when the controller is in the open position (e.g., when the controller is being used to control a UAV).

(37) In some implementations, as may be best seen in FIG. 9, a cross section C-D from FIG. 8, fins of the heatsink portion **172** may be interleaved with fins of the heatsink portion **174** with air gaps in between the fins. This may permit tightly fitting the heatsink portions **172** and **174** in the limited space available inside the controller **100**, while providing ease of access to components, such as the component **176** (e.g., comprising the communications circuitry).

(38) FIG. 13 is a flowchart of an example of a process **1300** for implementing a controller (e.g., the controller **100**) in which the controller includes a device support (e.g., the device support **110**) and a support stand (e.g., the support stand **120**). The process **1300** may include coupling **1310** a device support to a controller for a UAV in which the device support is configured to move between closed and open positions. The device support may be swingable, foldable, rotatable, and/or telescoping. The device support may be configured to hold a portable electronic device of various sizes, such as different sizes of a smart phone, and in some cases with an adapter, a mini tablet or a full size tablet (e.g., including sizes up to 10 inches in diameter). The device support may be movable between a closed position in which the device support is received in a bottom portion of the controller and an open position in which the device support extends away from the controller **100**. The closed position may be used when the controller is not in use, and the open position may be used when the controller is in use. In some implementations, a device support hinge (e.g., the device support hinge **118**) may be arranged between the controller and the device support for rotating the device support relative to the controller (e.g., from an open position to a closed position, and from a closed position to an open position). In the closed position, the device support may be received in the bottom portion of the controller. For example, the device support may fit in a central cavity of the bottom portion when in the closed position. In the open position, the device support may extend away from the bottom portion of the controller **100**.

(39) The process **1300** may also include coupling **1320** a support stand to the controller with the support stand being configured to move between open and open positions with the support stand. The support stand may be configured to support the controller at an angle relative to a surface, such as when the controller is resting on table, desk, or another platform. This may permit a user to use the controller, and use a device held by the controller, without the user physically holding the controller. The support stand may comprise individual support legs. The device support may be configured to cause the support stand to be received in the bottom portion of the controller when in the closed position. For example, the device support may pull the support legs of the support stand into side-cavities in the bottom portion. The device support may also be configured to permit the support stand to automatically extend away from the controller, such as to support the controller on a platform when in the open position. For example, the support legs may be spring loaded, and the device support may release the support legs from the side-cavities in the bottom portion when the device support **110** is moved to the open position.

(40) The process **1300** may also include integrating **1330** one or more antennas (e.g., the antennas **160A-160D**) in the device support and/or the support stand to communicate with a UAV. For example, the antennas may be connected to a component comprising communications circuitry that is arranged inside the controller. The antennas may be integrated in the device support, such as in an upper clamp portion (e.g., the upper clamp portion **144**). Moving the device support from the closed position to the open position may permit angling the antennas upward and/or outward in a direction overhead toward a UAV. Also, the antennas may be integrated in the support stand, such as in the support legs. Moving the device support from the closed position to the open position may also permit angling the antennas, integrated in the support stand, outward in a direction overhead toward a UAV when the controller is held by a user. Additionally, moving the device support from the open position to the closed position may improve protection of the antennas from environmental conditions, such as by pulling and/or locking the antennas (e.g., integrated in the device support) and the antennas (e.g., integrated in the support stand) in the bottom portion of the controller. Further, the antennas may be extended away from the controller when in the open position, into a fixed, known orientation, so that a user's hands do not contact the antennas or otherwise interfere with the antennas. Also, the antennas may be extended with horizontal separation and with vertical separation for improved communication with a UAV.

(41) FIG. **14** is a flowchart of an example of a process **1400** for implementing a controller (e.g., the controller **100**) in which the controller includes a fan (e.g., the fan **170**) and heatsink portions (e.g., heatsink portions **172** and **174**) arranged inside. The process **1400** may include arranging **1410** a fan in a controller for a UAV. The fan could be a blower fan configured to generate an airflow in the controller. For example, the fan may be configured to generate the airflow from an inlet, turn the airflow 90 degrees, force the airflow between the heatsink portions, and exhaust the airflow at an outlet, to move heated air out of the controller.

(42) The process **1400** may also include arranging **1420** heatsink portions in the controller with heatsink portions adjacent to one another and/or fins of the heatsink portions interleaved with air gaps in between. For example, the heatsink portions may be adjacent to one another, and the fan may be configured to generate an airflow between the heatsink portions adjacent to one another. In some implementations, fins of the heatsink portions (e.g., for dissipating heat) may be interleaved with one another with air gaps in between the fins. This may permit tightly fitting the heatsink portions in the limited space available inside the controller, while providing ease of access to components, such as a component comprising the communications circuitry. In some implementations, the heatsink portions may comprise a single heatsink. For example, the single heatsink may have an open channel in the middle to permit the airflow to pass therethrough. In some implementations, the heatsink portions may comprise multiple heatsinks. For example, the multiple heatsinks may be physically joined on sides with an open channel in the middle to permit the airflow to pass therethrough. In some implementations, the heatsink portions may include

thermal interface material (e.g., between a heatsink portion and a component).

(43) The process **1400** may also include configuring **1430** the fan to generate an airflow between the heatsink portions. The fan may be configured to generate the airflow between the heatsink portions that are adjacent to one another. For example, the airflow may ingress from an inlet arranged in a bottom portion of the controller, which inlet **184** may be exposed to ambient air when in the open position. Also, the airflow may egress (e.g., exhaust, exit, etc.) at an outlet arranged in the front portion of the controller, which outlet may also be exposed to ambient air when in the open position. The fan may generate the airflow from the inlet, turn the airflow 90 degrees, force the airflow between the heatsink portions, and exhaust the airflow at the outlet, to move heated air out of the controller. For example, the fan and the heatsink portions may use forced convection to dissipate heat from the components.

(44) While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law

Claims

1. An assembly, comprising: a controller for an unmanned aerial vehicle (UAV), the controller having a top portion and a bottom portion that defines a cavity; a device support coupled to the controller, the device support having a lower clamp portion and an upper clamp portion that are configured to hold a portable electronic device, wherein the device support is movable between a closed position, in which the device support is received in the cavity of the bottom portion and located adjacent to, and external from, the bottom portion such that the device support at least partially forms an exterior surface of the bottom portion, and an open position, in which the device support extends away from the controller; a support stand coupled to the controller, the support stand being configured to support the controller at an angle relative to a surface, wherein the support stand is spring loaded such that the device support pulls the support stand into the cavity of the bottom portion of the controller to position the support stand adjacent to the device support in the cavity of the bottom portion when in the closed position and permits the support stand to automatically extend away from the controller when in the open position; and antennas for communicating with a UAV, wherein the antennas include a first antenna integrated in the device support and a second antenna integrated in the support stand.
2. The assembly of claim 1, further comprising: a device support hinge between the controller and the device support, wherein the device support is configured to rotate between the closed position and the open position via the device support hinge.
3. The assembly of claim 1, wherein the first antenna is integrated in at least one of the lower clamp portion or the upper clamp portion and the second antenna is integrated in a first leg of the support stand, and wherein the assembly further comprises: a third antenna integrated in a second leg of the support stand.
4. The assembly of claim 1, further comprising: a release arranged on the controller, wherein the device support is configured to lock when in the open position, and wherein the release is configured to unlock the device support for moving the device support from the open position to the closed position.
5. The assembly of claim 1, wherein the support stand includes a support leg that includes a groove, and wherein a tab of the device support is configured to engage the groove to pull the support leg into the cavity of the bottom portion.
6. The assembly of claim 1, wherein the portable electronic device is configured to remain attached to the device support and located in the cavity of the bottom portion when the device support is in

the closed position.

7. The apparatus of claim 1, wherein the lower clamp portion and the upper clamp portion are configured on a spring loaded telescoping arm.

8. The apparatus of claim 1, further comprising: a fan arranged in the controller; and first and second heatsink portions configured to cool first and second components arranged in the controller, respectively, wherein the first and second heatsink portions are arranged adjacent to one another, and wherein the fan is configured to generate an airflow between the first and second heatsink portions.

9. The assembly of claim 1, further comprising: first and second heatsink portions configured to cool first and second components arranged in the controller, respectively, wherein the first and second heatsink portions are arranged adjacent to one another, and wherein fins of the first heatsink portion are interleaved with fins of the second heatsink portion with air gaps in between.

10. A system, comprising: a controller for a UAV, the controller having a top portion, a bottom portion, and a front portion extending between the top portion and the bottom portion, the front portion including a port; a device support coupled to the controller, the device support being configured to hold a portable electronic device for communicating with the UAV, wherein the device support is movable between a closed position, in which the device support is received in a recess of the bottom portion and the device support is positioned external to the bottom portion, and an open position, in which the device support extends away from the controller, and wherein the device support includes a channel that permits access to the port when the device support is in the closed position; a support stand coupled to the controller and configured to support the controller, wherein the support stand is configured for positioning adjacent to the device support in the recess of the bottom portion of the controller when the device support is in the closed position; and first and second heatsink portions configured to cool first and second components arranged in the controller, respectively, wherein the first and second heatsink portions are arranged adjacent to one another to permit an airflow between the first and second heatsink portions.

11. The system of claim 10, further comprising: a device support hinge between the controller and the device support, wherein the device support is configured to rotate between the closed position and the open position via the device support hinge.

12. The system of claim 10, further comprising: an antenna for communicating with the UAV, wherein the antenna is integrated in the device support.

13. The system of claim 10, further comprising: a support stand coupled to the controller, the support stand being configured to support the controller at an angle relative to a surface, wherein the device support causes the support stand to be received in the bottom portion of the controller when in the closed position and the device support permits the support stand to automatically extend away from the controller when in the open position.

14. The system of claim 10, further comprising: an antenna for communicating with the UAV; and a support stand coupled to the controller, the support stand being configured to support the controller at an angle relative to a surface, wherein the antenna is integrated in the support stand.

15. The system of claim 10, further comprising: a fan arranged in the controller wherein the fan is configured to generate an airflow between the first and second heatsink portions.

16. A method, comprising: coupling a device support to a controller for a UAV, the device support being configured to hold a portable electronic device, and the controller having a top portion, a bottom portion, and a front portion connecting the top portion and the bottom portion, the front portion having an outlet; coupling a support stand to the controller, the support stand being configured to support the controller at an angle relative to a surface; configuring first and second heatsink portions to cool first and second components arranged in the controller, respectively, wherein the first and second heatsink portions are arranged adjacent to one another to permit an airflow between the first and second heatsink portions, the airflow configured to exit the controller through the outlet of the front portion; and moving the device support between a closed position, in

which the device support and the support stand are received in the bottom portion of the controller, the device support and the support stand are located adjacent to one another, and at least a portion of the outlet of the top portion remains unobstructed by the device support, and an open position, in which the device support extends away from the controller, wherein the airflow is generated when the controller is in the open position, and wherein the portable electronic device is configured to remain secured by the device support when the device support is located in the closed position and the open position.

17. The method of claim 16, further comprising: configuring multiple antennas for communicating with a UAV, wherein the multiple antennas include a first antenna integrated in the device support and a second antenna integrated in the support stand.

18. The method of claim 16, further comprising: configuring a lower clamp portion and an upper clamp portion on the device support so that the lower clamp portion and the upper clamp portion hold a portable electronic device and so that the lower clamp portion and the upper clamp portion are external to the controller when in the closed position.

19. The method of claim 16, further comprising: configuring a spring with respect to the support stand, wherein the spring causes the support stand to extend away from the controller when the device support is in the open position.

20. The method of claim 16, further comprising: configuring a fan in the controller wherein the fan generates the airflow between the first and second heatsink portions.
